entangledECAR

Entangled cells in Reversible Elementary Cellular Automata

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1 What the program does

`entangledECAR` is a MATLAB script related to ECARdoku that generates visualisations of a reversible elementary cellular automaton (Rule 37R) in a *not fully determined regime*.

The program restricts the solution space one clue at a time. Instead of forcing a *unique* solution (as in classical Sudoku-style puzzles), it deliberately stops when exactly a small user-chosen number of solutions (typically 4–8) remain. At that point:

- The **determined** cells (always black or always white) have the same value in **all** retained solutions.
- The **indeterminate** cells take different values across the remaining solutions.
- Among the indeterminate cells, the program finds all **anticorrelated pairs** (pairs of cells whose values are always opposite in every remaining solution).
- Correlated indeterminate cells and fully determined cells are left uncoloured (grey, white, or black).
- Each anticorrelated pair is coloured with a distinct bright colour.

The result is a visual representation of apparent entanglement arising purely from global constraints in a perfectly classical, local, reversible cellular automaton.

2 How it works

1. **Precomputation (once)**: All 2^n possible initial rows are evolved for m generations using reversible Rule 37R (periodic boundaries + XOR carry term). The boards are stored as a 3-D `uint8` array.
2. **Interactive clue loop**: The user chooses a target number of solutions $N \geq 2$. Clues are added either manually or randomly until exactly N boards remain. The clues are saved in the text file `clues.txt`.
3. **Pair detection**: All pairs of indeterminate cells are examined. Only **anticorrelated** pairs are retained.
4. **Colouring**: Up to 8 anticorrelated pairs are selected and assigned distinct bright colours. All other cells stay white (0), grey (indeterminate), or black (1).
5. **Output**: A clean high-resolution PNG (100 pixels per cell) and the on-screen MATLAB figure are generated.

Important note on random mode: When using random clues, the program can occasionally enter a very long loop (or appear stuck) while searching for a set of clues that reduces the number of possible boards to exactly the requested target. This risk is higher for large target numbers (e.g. 100+). It usually does not happen when the chosen number of solutions is small (e.g. 4–12).

3 Usage

- Open `entangledECAR.m` in MATLAB.
- Run the script.
- Answer the on-screen prompts (manual or random clues, target number of solutions).
- The script will add clues, compute the remaining solutions, colour the anticorrelated pairs, and save the clean PNG.
- Repeat the target prompt as many times as you like (press Enter to exit).

4 Metaphysical musings

This program is an illustration of entanglement and of the deeper concept that the universe's choices (the clues) constrain what is possible. Choosing a value for one cell may instantly remove one of the possibilities for another cell — even one that is far away on the board — without any local interaction or signal passing between them. Or, two other cells may remain undetermined but become correlated or anticorrelated (entangled). Remarkably, a cell can be entangled not only with another cell in its present, but also with a cell in its future. This is clearly visible in the first row of the plots below (see Figure 1), where some cells are anticorrelated with cells several generations below them.

The same insight generalizes to a fully infinite horizontal grid. With any finite number of clues, or even with any countably infinite number of clues (unless they are arranged in very special ways, such as specifying the whole first row), the space of possible grids consistent with the clues remains uncountably infinite. This is somewhat equivalent to a Cantor set that remains uncountable even after removing a countably infinite set of uncountable sets. Indeterminacy cannot be exhausted.

A natural extension of the model is a growing grid in which two new indeterminate cells are added to the left and right edges at every time step. In this version the universe is literally expanding: at each new row fresh possibilities are created at the horizon while existing possibilities are continuously pruned by the clues. A clue given late in the simulation, even if given on one of the newly added edge cells, can retroactively force an anticorrelation with a cell that existed many generations earlier. It can also create a perfect anticorrelation between an early cell and a cell that is added much later. This makes the two distinct notions of time especially vivid: the internal *simulation time* (the vertical evolution of the grid) and the external *time of the simulating program* (the chronological order in which clues are successively given). The simulation can therefore entangle a cell with a cell in its own future.

The growing-grid construction eliminates the need to assume periodic boundaries.

This distinction has concrete consequences for simulation design. In the periodic case, specifying the last cell together with the first two cells of the first row fully determines the successor of the first cell due to wrap-around. In the growing-grid model this is not true, because the left neighbour of the first cell lies outside the tracked window and remains undetermined.

The non-locality is not merely in our knowledge of the system, but in the system itself. Once the clues are applied, the entangled pairs are still undetermined individually; the only

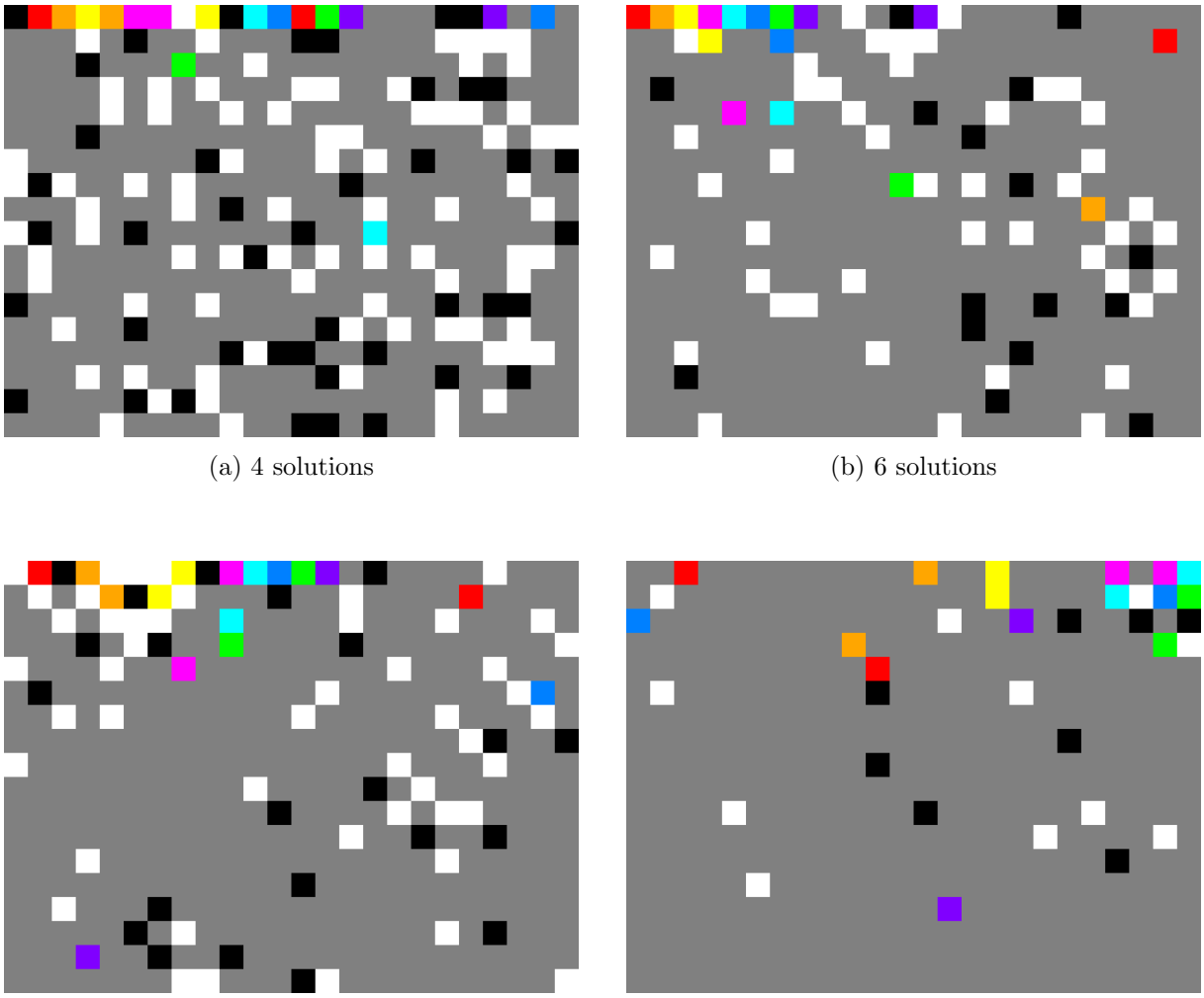
information that exists in the universe (that is, the set of all remaining consistent histories) is the entanglement relation itself. The non-locality is therefore a real property of the system.

The underlying physics (the local reversible rule) remains completely classical and local, yet the global structure of the possibility space that remains after the clues are applied exhibits objective non-local correlations.

This simple model reproduces some essential features of quantum mechanics: ontological uncertainty, and entanglement (without "spooky action at a distance"). It is also retrocausal in the sense that a future clue can determine a previous cell.

5 Example visualizations

The following figures show typical outputs for 24×24 boards (displayed cropped to 24×18 for readability) with 4, 6, 8 and 12 remaining solutions. Only anticorrelated pairs are highlighted with bright colours.



(a) 4 solutions

(b) 6 solutions

(c) 8 solutions

(d) 12 solutions

Figure 1: 24×24 boards (cropped to 24×18) showing only anticorrelated pairs in bright colours.